Results Math (2024) 79:30
© 2023 The Author(s)
1422-6383/24/010001-12
published online December 2, 2023
https://doi.org/10.1007/s00025-023-02061-8

Results in Mathematics



On a Coefficient Inequality for Carathéodory Functions

Paweł Zaprawa

Abstract. In the paper we consider a general inequality $|p_{n-1}p_{n+1}-{p_n}^2| \leq 4-|p_1|^2$ involving coefficients of functions with a positive real part. We prove this inequality for $n=2$ and $n=3$. Consequently, the relative inequalities involving coefficients of Schwarz functions are obtained. As an application, the two sharp estimates of the Hankel determinants $H_{3,1}$ and $H_{2,3}$ are proved for functions in $\mathcal{S}^*(1/2)$ and $\mathcal{M}$, respectively.

Mathematics Subject Classification. 30C50, 30C45.

Keywords. Carathéodory functions, Schwarz functions, coefficient problems, Hankel determinant.

1. Introduction

Denote by $\mathcal{P}$ the class of functions analytic in $\mathbb{D}=\{z\in\mathbb{C}:|z|<1\}$, given by

$$h(z)=1+\sum_{n=1}^{\infty}p_nz^n \tag{1}$$

and having a positive real part.

It is clear that

$$|p_{n-1}p_{n+1}-{p_n}^2|\leq 4 \tag{2}$$

is true for all $h\in\mathcal{P}$ and positive integers $n\geq 2$. It immediately follows from an inequality

$$|p_{n-1}p_{n+1}-{p_n}^2|\leq |p_{n-1}p_{n+1}-p_{2n}|+|p_{2n}-{p_n}^2|$$

and the inequality $|p_kp_{n-k}-p_n|\leq 2$ proved by Livingston ([10]). Equality in (2) holds for example for $h(z)=\frac{1+z^n}{1-z^n}$ which means that Inequlity (2) is sharp.

On the other hand, Inequality (2) is very often not strong enough to derive good results in many extremal problems concerning coefficients of analytic functions. For this reason we shall improve it by substituting the constant 4 by an expression depending on the first coefficient of h, i.e. $|p_1|$.

This idea appears in many papers concerning functions with positive real part. As an example it is worth recalling the result of Robertson ([12])

$$|p_{n+1} - p_n| \leq (2n+1)|2 - p_1|$$

or results obtained by Brown ([1]) and Lecko ([7]).

The functional $F(h) = |p_{n-1}p_{n+1} - {p_n}^2|$ defined for $h \in \mathcal{P}$ is rotationally invariant. It means that $F(h) = F(h_\varphi)$, where $h_\varphi(z) = h(e^{i\varphi}z)$. For this reason, if necessary, we can assume that $p_1 \in [0, 2]$.

To prove our results we need the following lemma (see, [6]).

Lemma 1. *If $p \in \mathcal{P}$ is of the form (1) with $p_1 \geq 0$, then*

$$2p_2 = p_1^2 + (4 - p_1^2)x, \tag{3}$$

$$4p_3 = p_1^3 + (4 - p_1^2)p_1x(2 - x) + 2(4 - p_1^2)(1 - |x|^2)y \tag{4}$$

and

$$\begin{aligned} 8p_4 = {} & p_1^4 + (4 - p_1^2)x\left[p_1^2(x^2 - 3x + 3) + 4x\right] \\ & - 4(4 - p_1^2)(1 - |x|^2)\left[p_1(x - 1)y + \overline{x}y^2 - \left(1 - |y|^2\right)w\right] \end{aligned} \tag{5}$$

for some x, y, $w \in \overline{\mathbb{D}} := \{z \in \mathbb{C} : |z| \leq 1\}$.

At the begining we shall estimate $|p_1p_3 - {p_2}^2|$ in terms of p_1.

Theorem 2. *If $h \in \mathcal{P}$ is given by (1), then*

$$|p_1p_3 - {p_2}^2| \leq 4 - |p_1|^2 \ . \tag{6}$$

Equality holds for rotations $h_t(\varepsilon z)$ and $g_t(\varepsilon z)$, $|\varepsilon| = 1$ of

$$h_t(z) = \frac{1 - z^2}{1 - 2tz + z^2} \ , \ t \in [-1, 1] \tag{7}$$

and

$$g_t(z) = \tfrac{1-t}{2}h_{-1}(z) + \tfrac{1+t}{2}h_1(z) = \frac{1 + 2tz + z^2}{1 - z^2} \ , \ t \in [-1, 1] \ . \tag{8}$$

Proof. From Lemma 1,

$$4\left(p_1p_3 - {p_2}^2\right) = (4 - {p_1}^2)\left[-4x^2 + 2p_1(1 - |x|^2)y\right].$$

Assuming $p = p_1 \in [0, 2]$ and writing $r = |x| \in [0, 1]$, we obtain

$$4\left|p_1p_3 - {p_2}^2\right| \leq (4 - p^2)\left[4r^2 + 2p(1 - r^2)\right] \leq (4 - p^2)\left[4r^2 + 4(1 - r^2)\right]$$

which results in (6). □

It is an easy exercise to show that equality in (6) holds for h_t and g_t defined by (7) and (8), respectively. In general case, if

$$|p_{n-1}p_{n+1} - {p_n}^2| \leq 4 - |p_1|^2 \tag{9}$$

holds true, then equality in (9) will be obtained also by h_t and g_t. Indeed, for h_t, $t \in [-1, 1]$ given by (7) we have

$$h_t(z) = 1 + \sum_{n=1}^{\infty} 2T_n(t)z^n \ , \tag{10}$$

where $T_n(t)$ stands for Chebyshev polynomials of the first kind. It is known for these polynomials that the inequality

$$T_{n-1}(t)T_{n+1}(t) - [T_n(t)]^2 = t^2 - 1$$

called Turan's inequality, holds for all integers $n \geq 2$. This results in the equality in (9).

Analogously, for g_t, $t \in [-1, 1]$ given by (8) there is

$$g_t(z) = 1 + 2tz + 2z^2 + 2tz^3 + \dots \ , \ t \in [-1, 1] \tag{11}$$

and so

$$p_{n-1}p_{n+1} - {p_n}^2 = 4(-1)^n(t^2 - 1) = (-1)^n({p_1}^2 - 4).$$

Unfortunately, we are not able to prove the conjecture (9), but we can do it in one more case, i.e. for $n = 3$.

2. Main Result

Theorem 3. *If $h \in \mathcal{P}$ is given by (1), then*

$$|p_2p_4 - {p_3}^2| \leq 4 - |p_1|^2 \ . \tag{12}$$

Equality holds for the same functions as in Theorem 2.

Proof. From Lemma 1, after tidious but elementary calculation,

$$\begin{aligned} 16\left(p_2p_4 - {p_3}^2\right) = (4 - {p_1}^2)\left[16x^3 - (1 - |x|^2)\left[16p_1xy + 4\left(4 - {p_1}^2 + {p_1}^2\overline{x}\right)y^2\right]\right. \\ \left. + 4(1 - |x|^2)(1 - |y|^2)\left[(4 - {p_1}^2)x + {p_1}^2\right]w\right]. \end{aligned}$$

From rotational invariance of $|p_2p_4 - {p_3}^2|$ we can assume that $p = p_1 \in [0, 2]$ and write $p = 2t$, $t \in [0, 1]$. Therefore,

$$\left|p_2p_4 - {p_3}^2\right| = (4 - p^2) \cdot |W| \ , \tag{13}$$

where

$$\begin{aligned} W = x^3 - (1 - |x|^2)\left[2txy + \left(1 - t^2 + t^2\overline{x}\right)y^2\right] \\ +(1 - |x|^2)(1 - |y|^2)\left[(1 - t^2)x + t^2\right]w. \end{aligned}$$

After rearranging terms and applying the triangle inequality we have

$$\begin{aligned}|W| &\leq \left|x^3-(1-|x|^2)\left(2txy+t^2\overline{x}y^2\right)\right| \\ &\quad +(1-|x|^2)(1-t^2)|y|^2+(1-|x|^2)(1-|y|^2)\left|(1-t^2)x+t^2\right|\end{aligned}$$

Let $x=re^{i\varphi}$, $y=\varrho e^{i\theta}$; hence

$$\begin{aligned}|W| &\leq \left|r^3e^{3i\varphi}-(1-r^2)\left[2tr\varrho e^{i(\varphi+\theta)}+t^2r\varrho^2e^{i(2\theta-\varphi)}\right]\right| \\ &\quad +(1-r^2)\varrho^2(1-t^2)+(1-r^2)(1-\varrho^2)\left|(1-t^2)re^{i\varphi}+t^2\right| \\ &\leq \left|e^{i(\varphi+\theta)}\left[r^3e^{i(2\varphi-\theta)}-2r(1-r^2)\varrho t-r(1-r^2)\varrho^2t^2e^{i(\theta-2\varphi)}\right]\right| \\ &\quad +(1-r^2)\varrho^2(1-t^2)+(1-r^2)(1-\varrho^2)\left[(1-t^2)r+t^2\right].\end{aligned}$$

Denote $\alpha=2\varphi-\theta$. The last inequality can be rewritten as follows

$$\begin{aligned}|W| &\leq r\left|r^2e^{i\alpha}-2(1-r^2)\varrho t-(1-r^2)\varrho^2t^2e^{-i\alpha}\right| \\ &\quad +(1-r^2)\varrho^2(1-t^2)+(1-r^2)(1-\varrho^2)\left[(1-t^2)r+t^2\right].\end{aligned}$$

Now, we shall maximize the expression

$$V=\left|r^2e^{i\alpha}-2(1-r^2)\varrho t-(1-r^2)\varrho^2t^2e^{-i\alpha}\right|$$

with respect to α. Consequently,

$$\begin{aligned}V^2 &= \left[[r^2-(1-r^2)\varrho^2t^2]\cos\alpha-2(1-r^2)\varrho t\right]^2+[r^2+(1-r^2)\varrho^2t^2]^2\sin^2\alpha \\ &= -4(1-r^2)\varrho tg(\cos\alpha)+r^4+(1-r^2)^2\varrho^4t^4+2r^2(1-r^2)\varrho^2t^2+4(1-r^2)^2\varrho^2t^2\end{aligned}$$

where

$$g(\tau)=r^2\varrho t\tau^2+\left(r^2-(1-r^2)\varrho^2t^2\right)\tau\quad \tau\in[-1,1].$$

Since $g'(\tau)=2r^2\varrho t\tau+\left(r^2-(1-r^2)\varrho^2t^2\right)$, we discuss three cases.

I. If $r\leq\frac{\varrho t}{1+\varrho t}$, then $g'(1)\leq 0$, so $g'(\tau)\leq 0$ for all $\tau\in[-1,1]$. Hence, g is a decreasing function of τ, so the lowest value of g is obtained for $\tau=1$. Consequently, the lowest values of V and W are obtained for $\alpha=0$. Moreover, for $\alpha=0$ there is

$$\begin{aligned}V &= \left|r^2-2(1-r^2)\varrho t-(1-r^2)\varrho^2t^2\right| \\ &= (1+\varrho t)^2\left|r^2-\frac{\varrho t(2+\varrho t)}{(1+\varrho t)^2}\right| \\ &= -r^2(1+\varrho t)^2+2\varrho t+\varrho^2t^2.\end{aligned}$$

For this reason,

$$|W|\leq -r^3+(1-r^2)h(\varrho,t)\equiv H(r,\varrho,t)$$

where

$$h(\varrho,t)=-2(1-r)\varrho^2t^2+(1-r)(\varrho^2+t^2)+2r\varrho t+r\quad \varrho,t\in[0,1].$$

The function h has at most one critical point (ϱ_0, t_0) inside $[0,1]\times[0,1]$ depending on r, where $\varrho_0 = t_0 = \sqrt{\frac{1}{2(1-r)}}$ and $h(\varrho_0, t_0) = \frac{1+2r-2r^2}{2(1-r)}$. For this critical point we have

$$H(\varrho_0, t_0) = \tfrac{1}{2}(1+3r-4r^3) \le 1$$

with equality for $r = 1/2$.

Considering values of h on the boundary of $[0,1]\times[0,1]$ results in

$$h(0,t) = r + t^2(1-r) \le 1$$
$$h(1,t) = -(1-r)t^2 + 2tr + 1 \le \frac{1-r+r^2}{1-r}$$

and so

$$H(0,t) \le -r^3 + (1-r^2)\cdot 1 \le 1$$
$$H(1,t) \le -r^3 + (1-r^2)\cdot\frac{1-r+r^2}{1-r} = 1$$

Since $h(\varrho,0) = h(0,\varrho)$ and $h(\varrho,1) = h(1,\varrho)$, thus $H(\varrho,0)$ and $h(\varrho,1)$ are also bounded by 1.

Summing up, if $r \le \frac{\varrho t}{1+\varrho t}$, then $|W| \le 1$. Observe that here and on it is not necessary to check whether the greatest value of particular functions discussed above are obtained for ϱ or t in the interval $[0,1]$.

II. If $r \ge \frac{\varrho t}{1-\varrho t}$, then $g'(-1) \ge 0$, so $g'(\tau) \ge 0$ for all $\tau \in [-1,1]$. Hence, g is a increasing function of τ, so the lowest value of g is obtained for $\tau = -1$. Consequently, the lowest values of V and W are obtained for $\alpha = \pi$. Moreover, for $\alpha = \pi$ there is

$$\begin{aligned} V &= \left|-r^2 - 2(1-r^2)\varrho t + (1-r^2)\varrho^2 t^2\right| \\ &= (1-\varrho t)^2\left|r^2 + \frac{\varrho t(2-\varrho t)}{(1-\varrho t)^2}\right| \\ &= r^2(1-\varrho t)^2 + 2\varrho t - \varrho^2 t^2. \end{aligned}$$

For this reason,

$$|W| \le r^3 + (1-r^2)h(\varrho,t) \equiv H(r,\varrho,t)$$

where

$$h(\varrho,t) = -2\varrho^2 t^2 + (1-r)(\varrho^2+t^2) + 2r\varrho t + r \quad \varrho, t \in [0,1].$$

The only critical point of the function h inside $[0,1]\times[0,1]$ is (ϱ_0, t_0), where $\varrho_0 = t_0 = \frac{\sqrt{2}}{2}$ and $h(\varrho_0, t_0) = \frac{1}{2} + r$. Additionally,

$$H(\varrho_0, t_0) = \tfrac{1}{2}(1+2r-r^2) \le 1$$

with equality for $r = 1$.

Considering values of h on the boundary of $[0,1]\times[0,1]$ results in

$$h(0,t)=r+t^2(1-r)\le 1$$
$$h(1,t)=-(1+r)t^2+2tr+1\le\frac{1+r+r^2}{1+r}$$

and so

$$H(0,t)\le r^3+(1-r^2)\cdot 1=1-r^2(1-r)\le 1$$
$$H(1,t)\le r^3+(1-r^2)\cdot\frac{1+r+r^2}{1+r}=1$$

Since $h(\varrho,0)=h(0,\varrho)$ and $h(\varrho,1)=h(1,\varrho)$, thus $H(\varrho,0)$ and $h(\varrho,1)$ are also bounded by 1.

Summing up, if $r\ge\frac{\varrho t}{1-\varrho t}$, then $|W|\le 1$.

III. Let $\frac{\varrho t}{1+\varrho t}<r<\frac{\varrho t}{1-\varrho t}$. In this case the lowest value of g is equal to

$$g(\tau_0)=-\frac{\left[r^2-(1-r^2)\varrho^2t^2\right]^2}{4r^2\varrho t}$$

where $\tau_0=\frac{(1-r^2)\varrho^2t^2-r^2}{2r^2\varrho t}$. Consequently,

$$V\le\frac{(1-r^2)\varrho^2t^2+r^2}{r}$$

and

$$|W|\le(1-r^2)\varrho^2t^2+r^2+(1-r^2)\varrho^2(1-t^2)+(1-r^2)(1-\varrho^2)\left[(1-t^2)r+t^2\right]$$
$$=1-(1-r)(1-r^2)(1-\varrho^2)(1-t^2)\le 1.$$

Combining cases I - III, we have $|W|\le 1$ for all $r,\varrho,t\in[0,1]$. This and (13) lead to (12). The sharpness of this result follows from the argument presented before Theorem 3. □

3. Relative Inequalities for Schwarz Functions

Let $\mathcal{B}_0$ be the class of Schwarz functions, i.e., analytic functions $\omega:\mathbb{D}\to\mathbb{D}$, $\omega(0)=0$, where $\mathbb{D}$ stands for the open unit disk $\{z\in\mathbb{C}:|z|<1\}$. The function $\omega\in\mathcal{B}_0$ can be written as a power series

$$\omega(z)=\sum_{n=1}^{\infty}c_nz^n\ ,\ z\in\mathbb{D}\ . \tag{14}$$

It is clear that if

$$p(z)=\frac{1+\omega(z)}{1-\omega(z)}$$

then

$$p\in\mathcal{P}\quad\text{if and only if}\quad\omega\in\mathcal{B}_0.$$

This property makes it possible to discuss problems in $\mathcal{B}_0$ considering the class $\mathcal{P}$ and vice versa. Further, we apply this property to establish a relation between the initial coefficients of $\omega \in \mathcal{B}_0$ and $p \in \mathcal{P}$.

Let $p(z)$ and $\omega(z)$ be of the form (1) and (14), respectively. Comparing coefficients at powers of z in

$$[1-\omega(z)]\,p(z) = 1+\omega(z)$$

we obtain

$$\begin{aligned} p_1 &= 2c_1 \\ p_2 &= 2c_2 + 2c_1^2 \\ p_3 &= 2c_3 + 4c_1c_2 + 2c_1^3 \\ p_4 &= 2c_4 + 4c_1c_3 + 2c_2^2 + 6c_1^2c_2 + 2c_1^4. \end{aligned} \tag{15}$$

Coefficient problems for functions $\omega \in \mathcal{B}_0$ were studied in numerous papers; for details, see ([2,3,11,13,15]). Here, we need the inequality obtained in [15].

Theorem 4. *If $\omega \in \mathcal{B}_0$ is given by (14), then*

$$\left|c_2c_4 - c_3^2\right| \le (1-|c_1|^2)^2\ . \tag{16}$$

Equality holds for rotations $\varepsilon^{-1}\omega(\varepsilon z)$, $|\varepsilon| = 1$ of

$$\omega(z) = \frac{z(t+z^2)}{1+tz^2} = tz + (1-t^2)z^3 - t(1-t^2)z^5 + \dots \quad t \in [-1,1].$$

Directly from Theorem 3 and (15) we get the following theorem.

Theorem 5. *If $\omega \in \mathcal{B}_0$ is given by (14), then*

$$|c_2c_4 - {c_3}^2 + {c_1}^2c_4 - 2c_1c_2c_3 + {c_2}^3| \le 1-|c_1|^2\ . \tag{17}$$

Equality holds for rotations $\varepsilon^{-1}\omega(\varepsilon z)$, $|\varepsilon| = 1$ of

$$\omega(z) = \frac{(t+z)z}{1+tz} = tz-(1-t^2)z^2-t(1-t^2)z^3-t^2(1-t^2)z^4+\dots\ ,\ t \in [-1,1]\ . \tag{18}$$

Finally, we can combine the inequalities from Theorem 5 and Theorem 4 to obtain more general inequality.

Theorem 6. *If $\omega \in \mathcal{B}_0$ is given by (14), then for all $\alpha \in [0,1]$*

$$|c_2c_4 - {c_3}^2 + \alpha\left({c_1}^2c_4 - 2c_1c_2c_3 + {c_2}^3\right)| \le (1-|c_1|^2)(1-(1-\alpha)|c_1|^2)\ . \tag{19}$$

4. Applications

As an application of the theorems presented in the two previous sections we derive the sharp bounds of Hankel determinants for two classes of analytic functions, namely for the class $\mathcal{S}^*(1/2)$ of starlike functions of order $1/2$ and the class $\mathcal{M}$ which consists of functions which are not necessarily univalent. Let us recall the definitions of these classes.

Denote by $\mathcal{S}^*(\alpha)$, $\alpha \in [0,1)$ the class of functions analytic in $\mathbb{D}$ such that

$$f(z) = z + \sum_{n=2}^{\infty} a_n z^n \tag{20}$$

and satisfying the condition

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) > \alpha.$$

In a similar way the class $\mathcal{M}(\alpha)$, $\alpha > 0$ is defined. Namely, this family consists of functions analytic in $\mathbb{D}$ with a power series expanion (20) such that

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) < 1 + \frac{\alpha}{2}.$$

Let $\mathcal{M} = \mathcal{M}(1)$. Clearly, if $f \in \mathcal{M}$, then

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) < \frac{3}{2}.$$

The Hankel determinants which we shall discuss are defined as follows:

$$H_{3,1} = a_3(a_2a_4 - {a_3}^2) + a_4(a_2a_3 - a_4) + a_5(a_3 - {a_2}^2) \tag{21}$$

and

$$H_{2,3} = a_3a_5 - {a_4}^2\ . \tag{22}$$

The problem of deriving bounds of Hankel determinants has focused the attention of many mathematicians in the recent years. It was solved for the majority of classes of analytic functions in case of the second Hankel determinant given by $H_{2,2} = a_2a_4 - {a_3}^2$. Generally speaking, this problem for $H_{3,1}$ is difficult, but today it is solved, among others, for main classes of function, for example for $\mathcal{S}^*$ or $\mathcal{K}$ (see, [4,5]). Usually, the calculation is very hard. The studies on $H_{2,3}$ are not so intensive as in case of $H_{3,1}$, only a few sharp results are known (see, [14]).

Theorem 7. *If $f \in \mathcal{S}^*(1/2)$ is given by (20), then*

$$|H_{2,3}| \leq \frac{3}{16}\ . \tag{23}$$

Equality holds for

$$f(z) = \frac{z}{\sqrt{1-z^2}} = z + \tfrac{1}{2}z^3 + \tfrac{3}{8}z^5 + \dots.$$

Proof. Each function $f \in \mathcal{S}^*(1/2)$ can be written in terms of $h \in \mathcal{P}$

$$\frac{zf'(z)}{f(z)} = \tfrac{1}{2}\left(1 + h(z)\right)$$

or equivalently, in terms of $\omega \in \mathcal{B}_0$

$$\frac{zf'(z)}{f(z)} = \frac{1}{1-\omega(z)}\ . \tag{24}$$

Hence, comparing coefficients in the equality $zf'(z)(1-\omega(z)) = f(z)$, where f and ω are of the form (20) and (14), respectively, we obtain

$$a_3 = \tfrac{1}{2}c_2 + {c_1}^2\ ,\ a_4 = \tfrac{1}{3}c_3 + \tfrac{7}{6}c_1c_2 + {c_1}^3\ ,\ a_5 = \tfrac{1}{4}c_4 + \tfrac{5}{6}c_1c_3 + \tfrac{3}{8}{c_2}^2 + \tfrac{23}{12}{c_1}^2c_2 + {c_1}^3\ . \tag{25}$$

Consequently,

$$\begin{aligned} H_{2,3} &= \frac{1}{144}\left(18c_2c_4 + 36{c_1}^2c_4 - 16c_3^2 + 24{c_1}^3c_3 - 52c_1c_2c_3\right.\\ &\quad \left. +27{c_2}^3 + 12{c_1}^4c_2 - 4{c_1}^2{c_2}^2\right)\\ &= \frac{1}{144}\left[16\left(c_2c_4 - c_3^2 + {c_1}^2c_4 - 2c_1c_2c_3 + {c_2}^3\right) + (2c_2 + 20{c_1}^2)c_4\right.\\ &\quad \left. + (24{c_1}^3 - 20c_1c_2)c_3 + 11{c_2}^3 + 12{c_1}^4c_2 - 4{c_1}^2{c_2}^2\right] \end{aligned}$$

and

$$|H_{2,3}| \le \frac{1}{144}G(|c_1|, |c_2|)$$

where

$$\begin{aligned} G(c,d) &= 16(1-c^2) + (2d + 20c^2)(1 - c^2 - d^2)\\ &\quad +(24c^3 + 20cd)\left(1 - c^2 - \tfrac{d^2}{1+c}\right) + 11d^3 + 12c^4d + 4c^2d^2\\ &= \tfrac{9-11c}{1+c}d^3 - \tfrac{8(2+5c)c^2}{1+c}d^2 + \left[2(1-c^2)(1+10c) + 12c^4\right]d\\ &\quad +4(1-c^2)(4 + 5c^2 + 6c^3). \end{aligned}$$

If $c \in [0, 9/11)$, then

$$\begin{aligned} G(c,d) &\le \tfrac{9-11c}{1+c}d^3 + \left[2(1-c^2)(1+10c) + 12c^4\right]d\\ &\quad +4(1-c^2)(4 + 5c^2 + 6c^3)\\ &\le \tfrac{9-11c}{1+c}(1-c^2)^3 + \left[2(1-c^2)(1+10c) + 12c^4\right](1-c^2)\\ &\quad +4(1-c^2)(4 + 5c^2 + 6c^3)\\ &= 27 - 7c^2 + 24c^3 - 19c^4 - 24c^5 - c^6. \end{aligned}$$

Denoting the latter by $g_1(c)$ we can see that

$$g_1'(c) = -2c(7 - 36c + 38c^2 + 60c^3 + 3c^4) \le 0 \quad \text{for all } c \in [0, 9/11)$$

and so in this case

$$G(c,d) \le g_1(0) = 27.$$

If $c \in [9/11, 1]$, then

$$\begin{aligned} G(c,d) &\leq \left[2(1-c^2)(1+10c)+12c^4\right] d + 4(1-c^2)(4+5c^2+6c^3) \\ &\leq \left[2(1-c^2)(1+10c)+12c^4\right] (1-c^2) + 4(1-c^2)(4+5c^2+6c^3) \\ &= 18+20c-16c^3-6c^4-4c^5-12c^6. \end{aligned}$$

Denoting the latter by $g_2(c)$ we obtain

$$g_2'(c) = 20-48c^2-24c^3-20c^4-72c^5 \leq 0 \quad \text{for all } c \in [9/11, 1]$$

and in this case

$$G(c,d) \leq g_2\left(\tfrac{9}{11}\right) < 18.$$

Combining both cases we have proven that $G(c,d) \leq 27$ and so $|H_{2,3}| \leq \frac{3}{16}$ with equality for $c=0$ and $d=1$. It means that the extremal function $f \in \mathcal{S}^*(1/2)$ is obtained when in (24) we put $\omega(z) = z^2$. □

Theorem 8. *If $f \in \mathcal{M}$ is given by (20), then*

$$|H_{3,1}| \leq \frac{3}{16}\,. \tag{26}$$

Equality holds for

$$f(z) = \frac{z}{\sqrt{1-z^2}} = z + \tfrac{1}{2}z^3 + \tfrac{3}{8}z^5 + \dots.$$

Proof. Each function $f \in \mathcal{M}$ can be written in terms of $h \in \mathcal{P}$

$$\frac{zf'(z)}{f(z)} = \tfrac{1}{2}\left(3-h(z)\right)$$

or equivalently, in terms of $\omega \in \mathcal{B}_0$

$$\frac{zf'(z)}{f(z)} = \frac{1-2\omega(z)}{1-\omega(z)}\,. \tag{27}$$

Hence, comparing coefficients in the equality $zf'(z)(1-\omega(z)) = f(z)(1-2\omega(z))$, where f and ω are of the form (20) and (14), respectively, we obtain

$$\begin{aligned} &a_2 = -c_1 \;\; a_3 = -\tfrac{1}{2}c_2 \;\; a_4 = -\tfrac{1}{3}c_3 - \tfrac{1}{6}c_1c_2 \\ &a_5 = -\tfrac{1}{4}c_4 - \tfrac{1}{6}c_1c_3 - \tfrac{1}{8}{c_2}^2 - \tfrac{1}{12}{c_1}^2c_2. \end{aligned} \tag{28}$$

Consequently,

$$\begin{aligned} H_{3,1} = \frac{1}{144}\big(&18c_2c_4 + 36{c_1}^2c_4 - 16c_3^2 + 24{c_1}^3c_3 \\ &-52c_1c_2c_3 + 27{c_2}^3 + 12{c_1}^4c_2 - 4{c_1}^2{c_2}^2\big)\,. \end{aligned}$$

Note that $H_{3,1}$ for $\mathcal{M}$ is exactly the same as $H_{2,3}$ for $\mathcal{S}^*(1/2)$. From the proof of Theorem 7 it follows our result. □

Funding This work was supported by FD-20/IM-5/140 from Lublin University of Technology.

Data availibility Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Declarations

Conflict of interest Author has no relevant financial or non-financial interests to disclose.

Open Access. This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

[1] Brown, J.E.: Successive coefficients of functions with positive real part. Int. J. Math. Anal. **4**, 2491–2499 (2020)

[2] Carlson, F.: Sur les coefficients d'une fonction bornée dans le cercle unité. Ark. Mat. Astr. Fys. **27A**, 1–8 (1940)

[3] Efraimidis, I.: A generalization of Livingston's coefficient inequalities for functions with positive real part. J. Math. Anal. Appl. **435**, 369–379 (2016)

[4] Kowalczyk, B., Lecko, A., Sim, Y.J.: The sharp bound for the hankel determinant of the third kind for convex functions Bull. Aust. Math. Soc. **97**, 435–445 (2018)

[5] Kowalczyk, B., Lecko, A., Thomas, D.K.: The sharp bound of the third Hankel determinant for starlike functions Forum Math. **34**, 1249–1254 (2022)

[6] Kwon, O.S., Lecko, A., Sim, Y.J.: On the fourth coefficient of functions in the Carathéodory class. Comput. Methods Funct. Theory **18**, 307–314 (2018)

[7] Lecko, A.: On coefficient inequalities in the Caratheodory class of functions. Ann. Pol. Math. **75**, 59–67 (2000)

[8] Libera, R.J., Złotkiewicz, E.J.: Coefficients bounds for the inverse of a function with derivative in $\mathcal{P}$. Proc. Amer. Math. Soc. **87**, 251–257 (1983)

[9] Libera, R.J., Złotkiewicz, E.J.: Early coefficients of the inverse of a regular convex function. Proc. Amer. Math. Soc. **85**, 225–230 (1982)

[10] Livingston, A.E.: The coefficients of multivalent close-to-convex functions. Proc. Amer. Math. Soc. **21**, 545–552 (1969)

[11] Prokhorov, D.V., Szynal, J.: Inverse coefficients for (α, β)-convex functions. Ann. Univ. Mariae Curie-Skłodowska Sect. A **35**, 125–143 (1981)

[12] Robertson, M.S.: Univalent functions starlike with respect to a boundary point. J. Math. Anal. **81**, 327–345 (1981)

[13] Schur, J.: Über Potenzreihen, die im Innern des Einheitskreises beschränkt sind. J. Reine Angew. Math. **147**, 205–232 (1917)

[14] Zaprawa, P.: On coefficient problems for functions starlike with respect to symmetric points. Bol. Soc. Mat. Mex. **28**, 17 (2022)

[15] Zaprawa, P.: Inequalities for the coefficients of Schwarz functions. Bull. Malays. Math. Sci. Soc. **46**, 144 (2023)

Paweł Zaprawa
Department of Mathematics
Lublin University of Technology
Nadbystrzycka 38D
20-618 Lublin
Poland
e-mail: p.zaprawa@pollub.pl

Received: July 12, 2023.
Accepted: October 31, 2023.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.